

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 4: Protocol Design Issues & Layered Architecture

Course: Computer Networks (Credit Hours: 3)

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1. PROTOCOL DESIGN ISSUES (RULES BANATE WAQT KIN CHEEZON KA KHAYAL RAKHNA HAI?)

Network ke protocols design karte waqt engineers ko kuch ahem maslon (design issues) ko hal karna padta hai:

- **Addressing (Pehchan):** Har computer ka ek unique pata (IP ya MAC address) hona zaroori hai taake data sahi receiver tak pahuche.
- **Error Control:** Raste mein signal kharab hone par galti pakadna (Error Detection) aur data dobara mangwana (Retransmission).
- **Flow Control:** Tez sender aur aahista receiver ki speed match karna taake receiver ka buffer memory bhar na jaye.
- **Routing:** Data bhejte waqt hazaron raston mein se sab se chhota aur behtareen rasta channa.

2. LAYERED ARCHITECTURE (LAYERS KYUN BANAYI JATI HAIN?)

Concept: Network ka pura kaam itna bada aur mushkil hai ke ek sath sambhalna namumkin hai. Is liye isko alag alag chhotay hisson (Layers) mein baant diya jata hai. Har layer ka apna ek makhsoos kaam hota hai.

Easy Example Courier Service: Aap sirf letter likh kar lifafay mein band karte hain. Courier boy use le jata hai, transport company gadi mein load karti hai. Har step ka banda apna alag farz nibhata hai.

3. OSI MODEL (7 LAYERS KA KHULASA)

Layer #	Layer Name	Asaan Alfaz Mein Kaam (Functions)
Layer 7	Application Layer	User interface jahan apps internet use karti hain (e.g., Browser, WhatsApp, Email, HTTP).
Layer 6	Presentation Layer	Data format convert karna, Encryption (security lagana), aur Data Compression.
Layer 5	Session Layer	Do computers ke darmiyan connection (session) banana, maintain rakhna aur close karna.
Layer 4	Transport Layer	End-to-End data deliver karna, data ke tukray (Segments) banana, Flow Control aur Error Control (TCP/UDP).
Layer 3	Network Layer	Logical Addressing (IP Address lagana) aur behtareen rasta dhoondna (Routing). Data unit: Packet .
Layer 2	Data Link Layer	Physical Addressing (MAC Address lagana), Hop-to-Hop delivery. Data unit: Frame .
Layer 1	Physical Layer	Asal taarein, radio waves, bits (0 aur 1) ko voltage ya roshni mein bhejti hai.

4. QUICK STUDENT REVIEW QUESTIONS

Q1: Flow Control aur Error Control mein kya farq hai?

Ans: Flow Control sender aur receiver ki speed ko barabar rakhta hai taake receiver dab na jaye, jabke Error Control kharab hue data ko pakad kar dobara mangwata hai.

Q2: 7 Layers yaad rakhne ka aasan tareeqa kya hai?

Ans: Jumla yaad rakhein: **"Please Do Not Throw Sausage Pizza Away"** (Physical, Data Link, Network, Transport, Session, Presentation, Application).